

Angular HandsOn 4: Template Driven Forms and Validation

HANDS-ON 4 [Intermediate]

Template-Driven Forms & Validation

Topics Covered:

Template-Driven Form Basics[Check] ngModel Two-Way Binding in Forms[Check]

Built-in Validators ? required, minlength, [Check]

email

ngForm and ngModel References[Check]

Displaying Validation Errors[Check] Form Submission Handling[Check]

Template-driven forms are Angular's simpler, HTML-centric form approach. You will build a Student Enrollment Request form using template-driven forms, add built-in validation, display contextual error messages, and handle form submission.

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Add FormsModule to the component's imports array (standalone) or the module's imports to enable ngModel and ngForm.

Task 1: Build the Enrollment Request Form

Goal: Create a template-driven form with multiple input types and ngModel bindings.

Steps:

38. Generate an EnrollmentFormComponent: ng generate component pages/enrollment-form. Add it as a route in app-routing.module.ts at path /enroll.

39. In the template, build a form using <form #enrollForm='ngForm' (ngSubmit)='onSubmit(enrollForm)'>. Add the following fields, each with [(ngModel)] and a name attribute: studentName (text input), studentEmail (email input), courseId (number input), preferredSemester (select with options: Odd, Even), agreeToTerms (checkbox).

40. In the component class, define an onSubmit(form: NgForm) method that logs form.value and form.valid to the console.

41. Add a Submit button: <button type='submit' [disabled]='enrollForm.invalid'>Submit</button>.

42. Test the form: submit with all fields filled and observe the logged values. Note the structure of form.value.

Hint:

- The #enrollForm='ngForm' template reference variable gives you access to the NgForm directive instance ? use it to check form validity, reset the form, or access individual control states.

- name attribute is mandatory on every form control when using template-driven forms ? Angular uses it as the key in form.value.

Expected Outcome: Console logs the form value object with all field names as keys on submit. Submit button is disabled when the form is invalid.

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Task 2: Validation and Error Messages

Goal: Add built-in validators and display contextual error messages for each field.

Steps:

43. Add validation attributes to studentName: required and minlength='3'. Add required and email to studentEmail. Add required to courseId. Add required (Checked) to agreeToTerms.

44. For studentName, display error messages conditionally: show 'Name is required' when the control is touched and has the required error; show 'Name must be at least 3 characters' when it has the minlength error. Use the pattern: `<span *ngIf='nameCtrl.touched && nameCtrl.errors?.required'>.</span>` after adding `#nameCtrl='ngModel'` to the input.

45. Apply Angular's form state CSS classes to style inputs: add CSS rules for `.ng-invalid.ng-touched` { border-color: red; } and `.ng-valid.ng-touched` { border-color: green; } in the component's stylesheet.

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46. Add a success message `<div>` that appears after successful submission: 'Enrollment request submitted successfully!' Use a submitted boolean property toggled in `onSubmit`, and `*ngIf` to show/hide it.

47. Add a Reset button that calls `enrollForm.resetForm()` to clear all fields and validation states.

Hint:

- `touched` is true once the user has focused and left the field. `dirty` is true once the user has changed the value. Use `touched` for showing errors ? not `dirty` ? since errors show after the user interacts with a field, not while typing.

- Angular automatically adds CSS classes to form controls: `ng-valid/ng-invalid` (validation state), `ng-pristine/ng-dirty` (changed state), `ng-touched/ng-untouched` (visited state).

Expected Outcome: Invalid fields show red borders and specific error messages after being touched. Valid fields show green borders. Successful submission shows the success message.

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